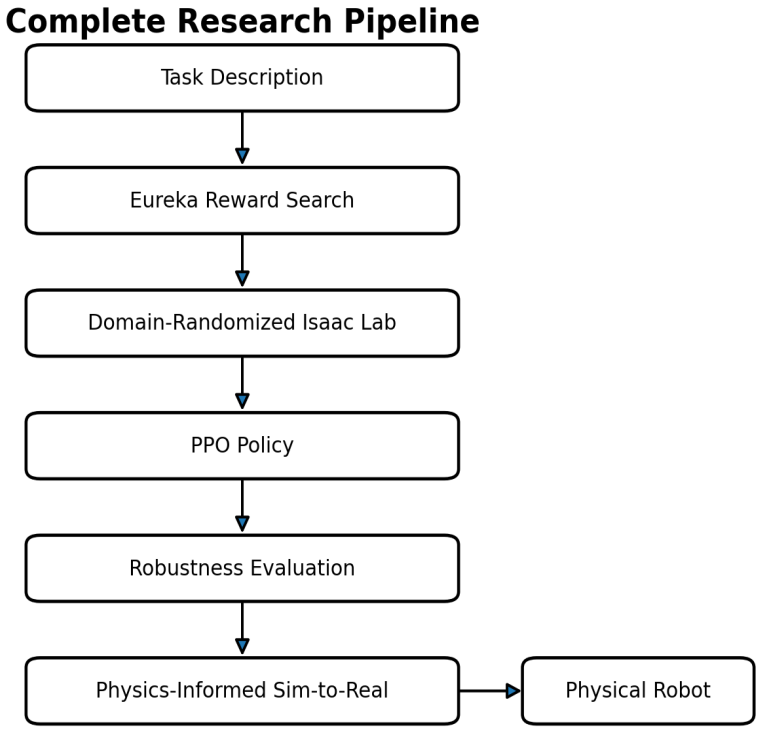
2-DOF PLANAR ROBOTIC REACHING

Complete Technical Research Report

PPO • Human Reward Engineering • Eureka Reward Discovery • Domain Randomization • Future PINN-Based Sim-to-Real



Scope. This report reconstructs and expands the complete work documented in the supplied 53-page technical review, preserving its experimental findings, mathematical formulation, configuration, limitations, and proposed next stages. It distinguishes completed experiments from proposed future work.

Source basis: supplied “2-DOF Robotic Arm Technical Review.pdf”

Compiled: 15 September 2026

Executive Summary

The project studies automated reward engineering for reinforcement learning in a deliberately controlled robotics problem: a two-link, two-revolute-joint planar arm in Isaac Lab that must move the tip of its second link to randomly generated Cartesian targets. The policy is trained with PPO. The central experimental variable is not PPO itself, but how the reward is produced: either manually by a human or automatically by Eureka using an LLM.

The experimental design is scientifically useful because the robot, observation space, action space, physics, target distribution, PPO architecture, and evaluation procedure are kept fixed as far as possible. This isolates reward generation as the main difference between the two systems.

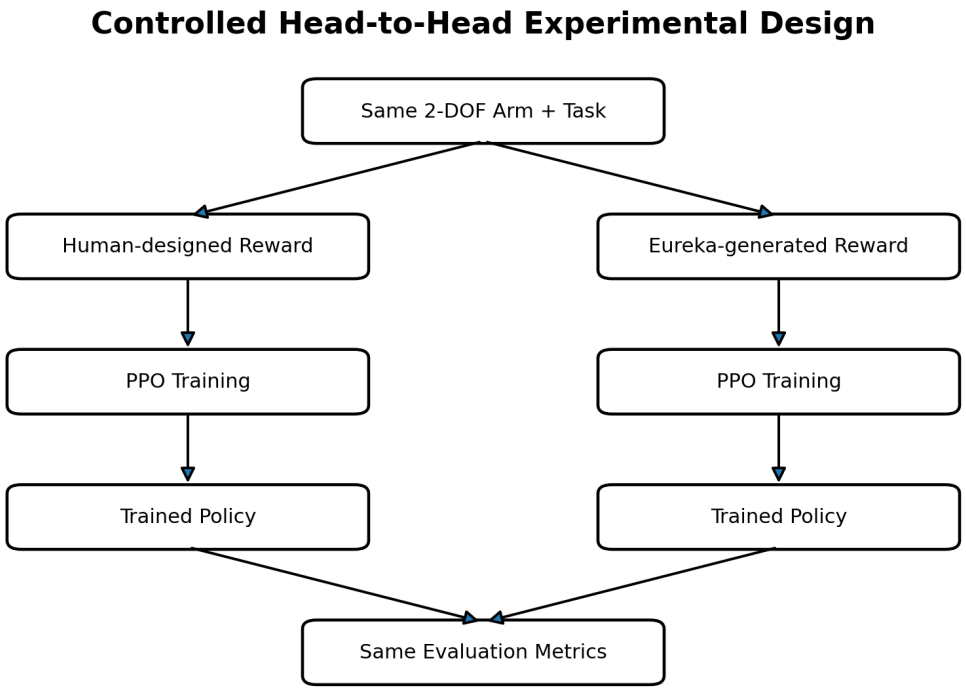


Figure 1. Controlled comparison between the human-designed and Eureka-generated reward pipelines.

The human-designed baseline reward is dense and precision-friendly: it continuously rewards smaller Cartesian distance, adds a 5 cm success bonus, and penalizes squared normalized action magnitude. After 5,000 PPO iterations, the baseline achieved 99.49% success within 5 cm, 98.36% within 2 cm, 79.74% within 1 cm, 7.6 mm mean terminal error, approximately 0.315 rad/s mean joint velocity, and approximately 0.218 control effort.

Eureka, using Gemini, discovered a qualitatively different reward containing exponential proximity, a very large 5 cm success bonus, a joint-velocity penalty, and a per-step time penalty. At 500 PPO iterations it reached 97.90% success within 5 cm, but only 19.02% within 2 cm and 3.56% within 1 cm. Its mean terminal error was 33.5 mm, mean joint velocity 4.84 rad/s, and mean control effort about 2.20.

> The most important interpretation is therefore not “Eureka beats PPO.” PPO is used in both cases. Eureka operates one level above PPO by changing the objective PPO optimizes. The discovered reward rapidly solved the coarse reaching objective but strongly incentivized crossing the 5 cm boundary and did not provide a sufficiently strong economic reason to refine a successful state from centimeters toward millimeters.

A later 1,500-iteration Eureka run reinforced this conclusion: the policy remained around 95.3% at 5 cm, 18.1% at 2 cm, and 5.3% at 1 cm, with approximately 45.5 mm mean terminal error, 4.79 rad/s mean joint velocity, and 2.54 mean control effort. The source document therefore concludes that more PPO training did not repair a reward whose objective itself was poorly aligned with fine precision.

Table 1. Core documented evaluation results.

Metric	Human / Vanilla PPO	Eureka PPO (500 it.)	Eureka PPO (1,500 it.)
PPO iterations	5,000	500	≈1,500
Success < 5 cm	99.49%	97.90%	95.3%
Success < 2 cm	98.36%	19.02%	18.1%
Success < 1 cm	79.74%	3.56%	5.3%
Mean terminal error	7.6 mm	33.5 mm	45.5 mm
Median terminal error	6.3 mm	34.6 mm	33.3 mm
Mean joint velocity	0.315 rad/s	4.84 rad/s	4.79 rad/s
Mean control effort	≈0.218	≈2.20	≈2.54

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1. Project Objective and Research Question

1.1 Fundamental research question

The original question is whether an LLM/Eureka can discover a reward function that enables PPO to learn the 2-DOF reaching task effectively, potentially matching or outperforming a manually designed human reward. The experiment deliberately separates reward engineering from policy optimization.

1.2 Eureka versus PPO

PPO answers the policy-learning problem: given a reward, what policy should be learned to maximize expected discounted return? Eureka answers a different problem: what reward function should be given to the RL system? Thus the conceptual hierarchy is reward engineering → Eureka/LLM reward search → PPO → neural policy → robotic behavior.

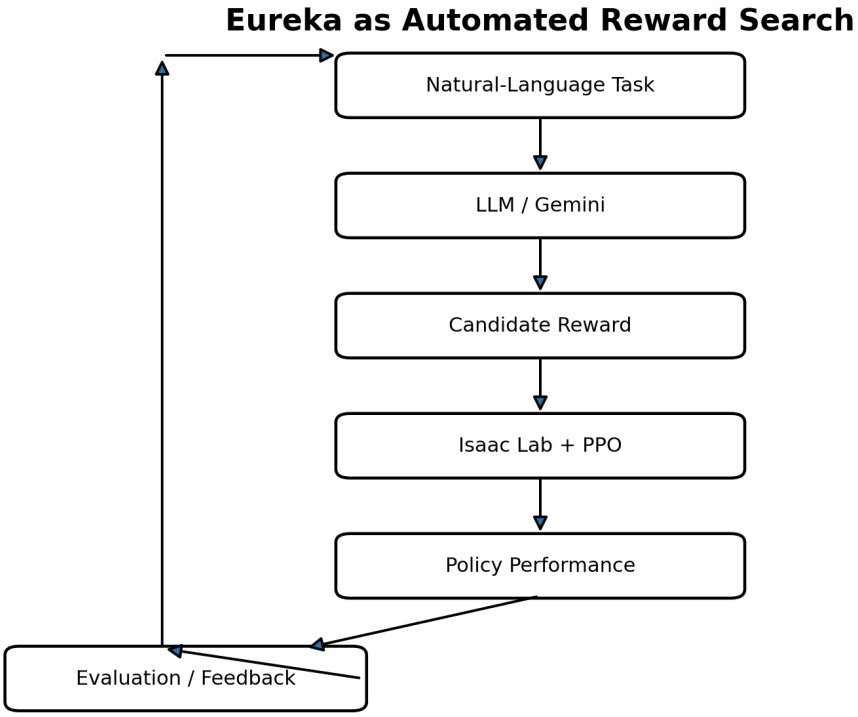


Figure 2. Eureka is an outer reward-search loop around the RL training process.

1.3 What the project is not claiming

It is incorrect to describe Eureka as a better PPO algorithm or as a replacement for PPO. The actual experimental manipulation is the objective presented to PPO. This distinction is essential when interpreting the observed speed-versus-precision tradeoff.

2. Why the 2-DOF Experimental Platform

2.1 Deliberate simplification

The project intentionally excludes grasping, object manipulation, wrist orientation, contact-rich manipulation, dexterous hand control, and object placement. The sole task is to move the tip of the second link to a randomly generated target in the XY plane.

2.2 Why simplicity is scientifically useful

With a complicated robot, a failure could originate from dynamics, observations, actions, PPO, reward engineering, contact dynamics, or Eureka. A two-joint planar arm makes the central variable—reward design—much easier to isolate.

2.3 Task statement

At each reset, a new target is generated. The controller therefore cannot simply memorize one target. It must learn a general mapping from joint state, target information, and Cartesian error to torque commands.

3. Robot Architecture and Physical Parameters

3.1 Serial structure

The robot is a planar serial manipulator consisting of a fixed base, joint 1, link 1, joint 2, link 2, and an effective end-effector at the tip of link 2. There is no physical gripper and no separate end-effector link.

3.2 Geometry and mass

Both links are 0.5 m long. Link 1 has a documented mass of 1.0 kg and link 2 a mass of 0.5 kg. Both joints rotate around the Z axis, producing planar motion in XY. The maximum straight-line reach is approximately 1.0 m.

3.3 Joint limits and nominal configuration

The documented joint position range is approximately q_i ∈ [–π, π], with torque/effort limits of ±20 Nm and velocity limits of approximately ±5 rad/s. The nominal initial joint state is q_1=q_2=0 with zero velocity, while episode resets randomize joint positions over [–π,π] and reset velocities to zero.

3.4 Damping/friction

The supplied review documents a nominal friction/damping coefficient of approximately 0.05 for each joint and a physics-level integration performed by Isaac Lab/PhysX.

4. Mathematical Model and Task Definition

4.1 Manipulator dynamics

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) + F(\dot{q}) = \tau + \tau_{\text{ext}}$$

Equation 1. Standard two-link manipulator dynamics used as the conceptual physical model.

Interpretation

Here q is the joint-position vector, q■ and q■ are joint velocity and acceleration, M(q) is the configuration-dependent inertia matrix, C(q,q■)q■ represents Coriolis/centrifugal effects, g(q) gravity, F(q■) friction/damping, τ commanded actuator torque, and τ_ext external disturbance torque/force effects. The policy does not explicitly solve this equation; PhysX performs the numerical integration and PPO learns a policy that implicitly controls it.

4.2 Forward kinematics

$$x_{ee} = l_1\cos q_1 + l_2\cos(q_1+q_2), \qquad y_{ee} = l_1\sin q_1 + l_2\sin(q_1+q_2)$$

Equation 2. Forward kinematics from joint space to the end-effector position.

With l■=l■=0.5 m

The end-effector coordinates become the direct function of q■ and q■. This is the mathematical bridge between the low-level joint controller and the Cartesian reaching objective.

4.3 Target generation

$$r \sim U(0.20, 0.85), \quad \theta \sim U(0, 2\pi), \quad x_t = r\cos\theta, \quad y_t = r\sin\theta$$

Equation 3. Random target sampling in polar coordinates and conversion to Cartesian coordinates.

Interpretation

The radius is sampled from 0.20 to 0.85 m and the angle uniformly from 0 to 2π. The target therefore covers a broad portion of the reachable workspace while remaining inside the approximately 1 m maximum reach.

4.4 Cartesian error

$$e_t = p_t - p_{ee,t}, \qquad d_t = \|e_t\|_2 = \sqrt{(x_t - x_{ee,t})^2 + (y_t - y_{ee,t})^2}$$

Equation 4. Cartesian target error and Euclidean terminal distance.

Core control objective

The scalar d is the main quantity used by both reward functions and the evaluation system. The ultimate geometric goal is d → 0.

5. Observation, Action, Simulation, and Parallelization

5.1 Observation space

$$s_t = [q_1, q_2, \dot{q}_1, \dot{q}_2, x_t, y_t, x_t - x_{ee}, y_t - y_{ee}]$$

Equation 5. Eight-dimensional policy observation.

The eight values are q■, q■, q■■, q■■, x_t, y_t, x_t–x_ee, and y_t–y_ee. The observation therefore includes both absolute target information and relative target-to-end-effector information.

5.2 Torque action model

$$\tau_i = 20a_i, \qquad a_i \in [-1, 1], \qquad \tau_i \in [-20, 20]\ \text{Nm}$$

Equation 6. Normalized action to physical torque mapping.

Examples

a_i=1 gives 20 Nm, a_i=0.5 gives 10 Nm, a_i=0 gives zero torque, and a_i=–1 gives –20 Nm. This is genuine low-level effort control rather than direct joint-position control.

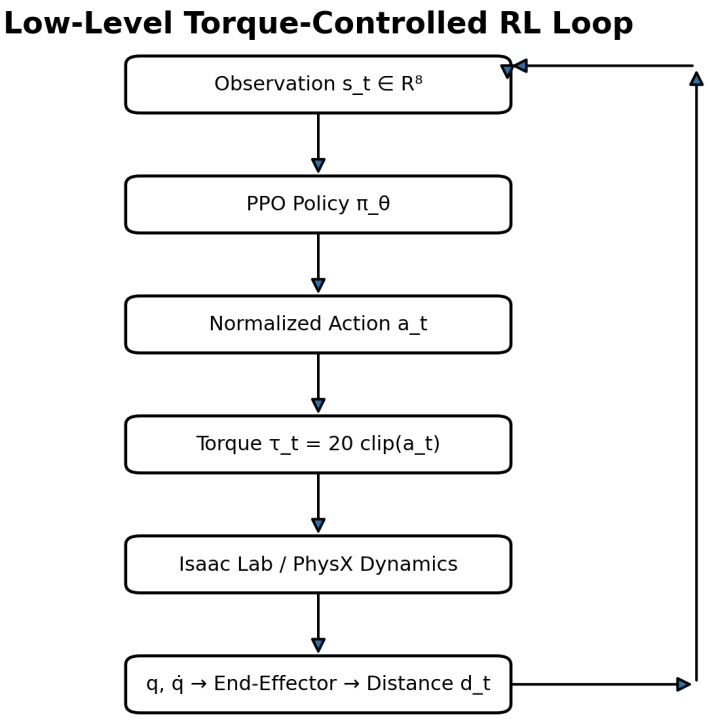


Figure 3. Observation-to-torque-to-dynamics control loop.

5.3 Simulation and control rates

The physics timestep is 1/120 s, so PhysX runs at 120 Hz. Control decimation is 4, giving a policy frequency of 30 Hz. Each policy action is held for four physics integration steps.

5.4 Episode duration

Episodes last 10 s. At 30 policy decisions per second, this corresponds to approximately 300 policy decisions per episode. At 120 Hz physics, there are approximately 1,200 physics integration steps per episode.

5.5 Massive parallelization

The environment uses 4,096 parallel simulated environments. A single rollout therefore contains 4,096 × 24 = 98,304 transitions when 24 steps are collected per environment before a PPO update. This parallelism is a major reason the experiments can gather large numbers of trajectories efficiently.

6. PPO Architecture and Training Configuration

6.1 Actor and critic network

The documented actor/critic backbone is 8 → 256 → 128 → 64, with ELU activations. The actor outputs a Gaussian action distribution over the two normalized torque actions. The critic estimates V(s), the expected future return.

6.2 Gaussian policy

The action components are modeled conceptually as a■~N(μ■,σ■) and a■~N(μ■,σ■), with initial standard deviation σ=1.0. Exploration therefore starts broad and becomes more concentrated as learning progresses.

6.3 PPO hyperparameters

Parameter	Documented value
Learning rate	1×10■³
Discount γ	0.99
GAE λ	0.95
Clip parameter ε	0.2
Value-loss coefficient	1.0
Entropy coefficient	0.005
Desired KL	0.01
Max gradient norm	1.0
Learning epochs	8
Mini-batches per rollout	4

Table 2. Consolidated PPO configuration from the supplied review.

6.4 PPO objective

At a high level, PPO searches for a policy maximizing expected discounted return under the supplied reward. The important point for this project is that PPO's optimization target changes when Eureka changes the reward.

7. Human-Designed Reward and Vanilla Baseline

7.1 Reward definition

$$R_{\text{vanilla}} = -d + 5\,\mathbb{I}(d < 0.05) - 0.005\|a\|_2^2$$

Equation 7. Human-designed baseline reward.

7.2 Distance shaping

The −d term is dense shaping. If d=0.5 m, the distance contribution is −0.5; if d=0.2 m it is −0.2; if d=0.05 m it is −0.05. Thus getting closer remains continuously better.

7.3 Success bonus

The indicator is 1 when d<0.05 m and zero otherwise, giving a +5 reward for entering the 5 cm success region.

7.4 Action regularization

The −0.005||a||² term discourages unnecessarily large normalized torque commands. Because the penalty is quadratic, larger actions are penalized more strongly than small actions.

7.5 What the human reward implicitly communicates

The reward tells PPO to move toward the target, enter the 5 cm region, and avoid excessive control effort. It does not explicitly include a settling-time term, explicit joint-velocity penalty, or overshoot penalty. Nevertheless, the continuous −d term keeps providing an incentive to improve precision even after entering the success region.

7.6 Baseline training

The vanilla policy was trained for 5,000 iterations. The documented wall-clock duration was approximately 5,514.49 s (about 91.9 minutes). Late training logs showed mean reward around 1,400–1,420, episode length 300 steps, and action standard deviation around 0.10–0.11. The absolute reward magnitude should not be compared directly with Eureka's return.

7.7 Baseline performance

Metric	Vanilla PPO
Success < 5 cm	99.49%
Success < 2 cm	98.36%
Success < 1 cm	79.74%
Mean terminal error	7.6 mm
Median terminal error	6.3 mm
Mean joint velocity	0.315 rad/s
Mean control effort	≈0.218

Table 3. Documented final vanilla evaluation.

8. Eureka Reward Discovery

8.1 Eureka's role

Eureka is an automated reward-design/search mechanism. It receives a task description and environment interface, generates candidate reward functions, trains policies using those rewards, evaluates policy performance, and feeds the measured result back into the search process.

8.2 Task description

The task was described approximately as moving the 2-DOF arm's end-effector to the target quickly and accurately. The success metric was centered on target distance with a tolerance corresponding to the 5 cm task threshold.

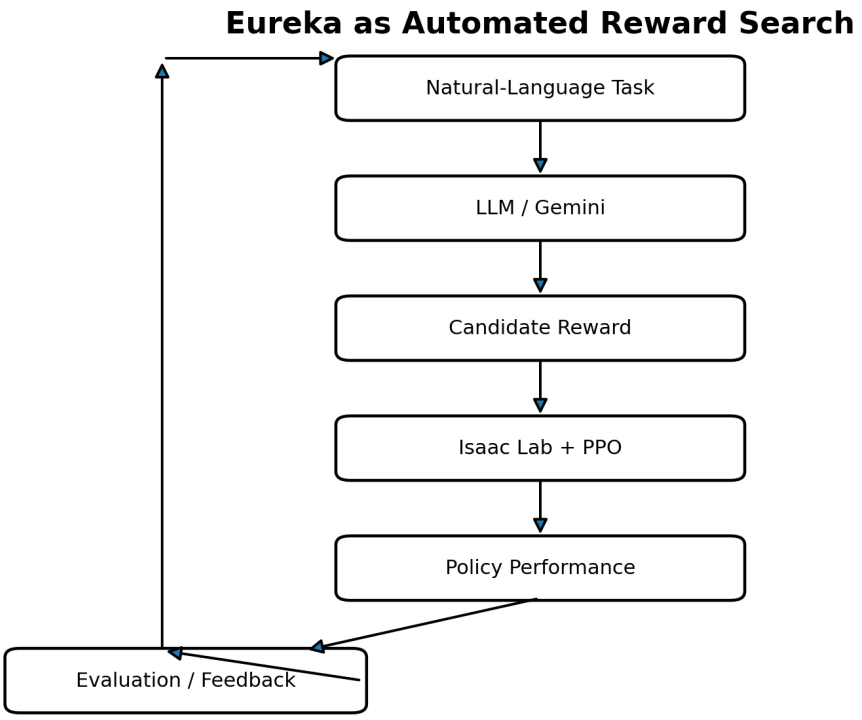


Figure 4. Reward discovery loop documented in the project.

8.3 Discovered reward

$$R_{\text{Eureka}} = e^{-10d} + 100\,\mathbb{I}(d < 0.05) - 0.01\|\dot{q}\|_2^2 - 0.1$$

Equation 8. Eureka-generated reward documented in the project.

8.4 Exponential proximity

The e^(–10d) term is a nonlinear proximity reward. It approaches 1 at d=0, is approximately 0.6065 at d=5 cm, and is approximately 0.0067 at d=0.5 m. It therefore gives useful distance shaping, but its absolute scale remains small relative to the +100 success bonus.

8.5 Dominant success bonus

The Eureka reward gives +100 whenever d<0.05 m, versus +5 in the human reward. The success bonus is therefore 20 times larger. Because a 10-second episode has approximately 300 policy steps, sustained success-region occupancy can produce thousands of reward points.

8.6 Time penalty

The –0.1 term penalizes every step. This creates explicit pressure to finish quickly, unlike the human reward, which does not contain a direct per-step time cost.

8.7 Velocity penalty

The –0.01||q■||² term penalizes physical joint velocity rather than normalized action magnitude. This is a meaningful change in regularization: large commands are no longer the direct object of the penalty; high physical motion is.

8.8 Behavioral implication

Together, the time penalty and enormous success bonus create urgency. Outside the 5 cm region, every step incurs a time cost. Crossing the threshold produces a very large positive jump. The learned controller consequently becomes more aggressive, with much higher joint velocity and control effort.

9. Head-to-Head Results and Interpretation

9.1 First exact comparison

The first head-to-head evaluation used identical conditions: the same environment, 4,096 parallel PhysX environments, the same 300-step evaluation horizon, the same target setup, and the same measurement methodology.

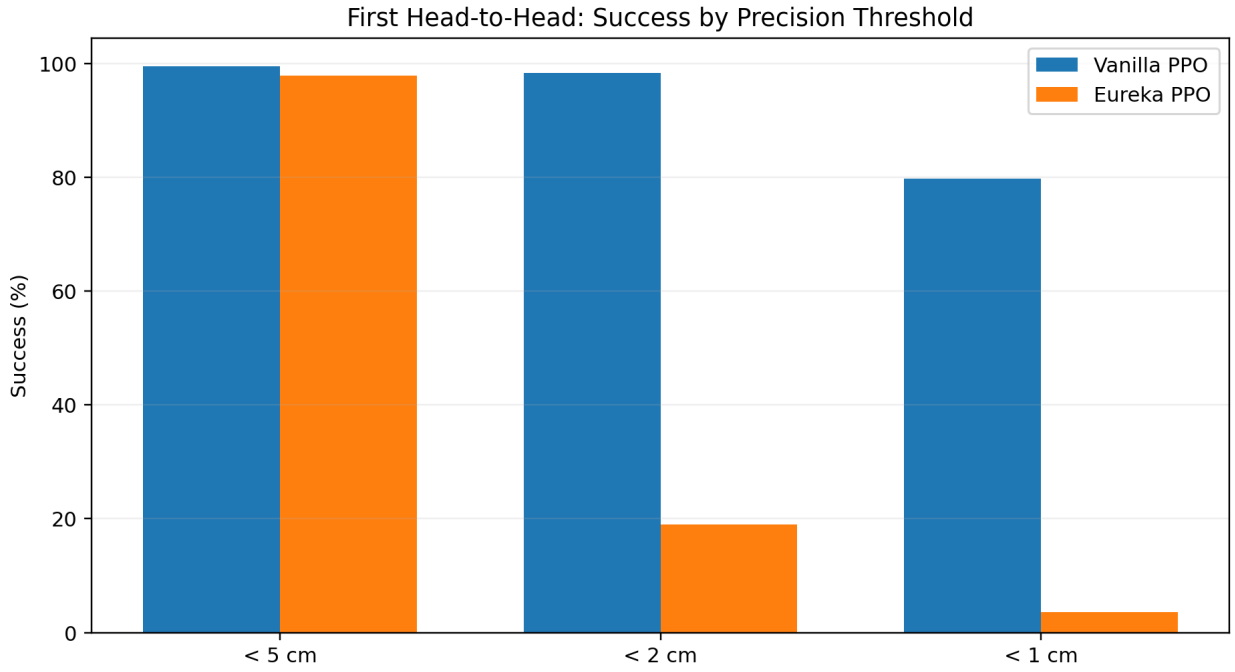


Figure 5. The major precision gap is hidden if only 5 cm success is reported.

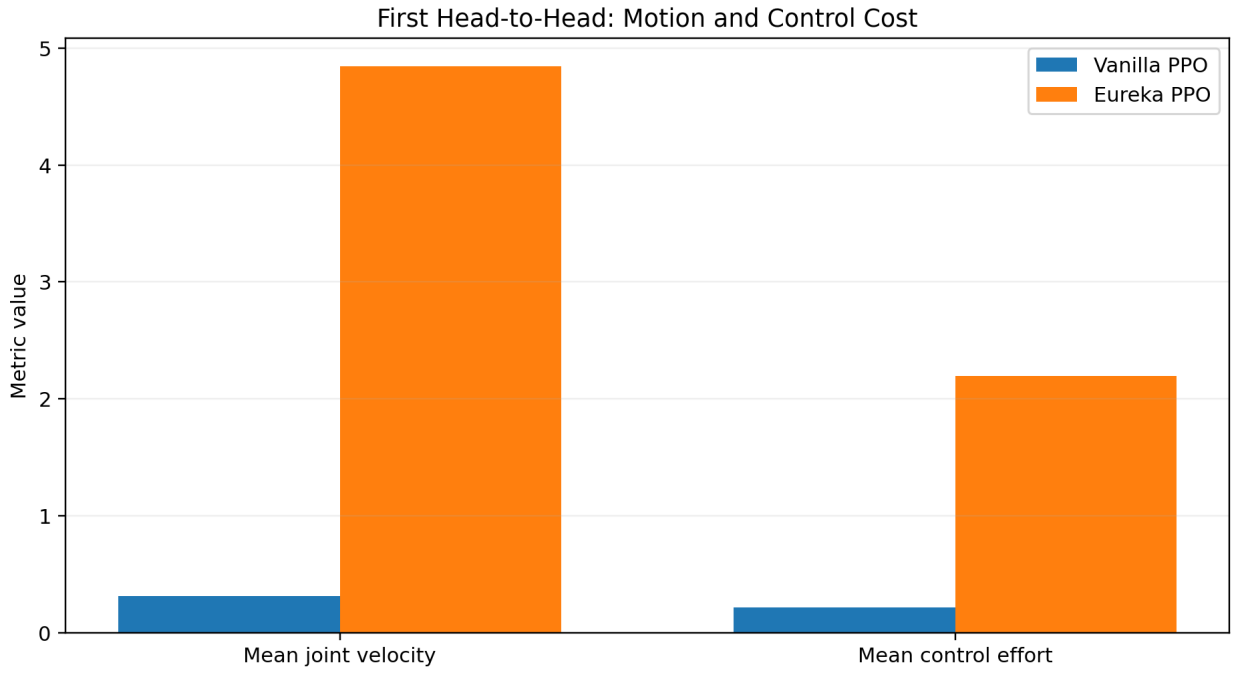


Figure 6. Eureka's early policy is much more aggressive in velocity and control effort.

9.2 Speed of useful learning

Eureka reached 97.90% success within 5 cm after 500 PPO iterations, while the vanilla baseline reached 99.49% after 5,000 iterations. The documented comparison is therefore approximately a 10× difference in PPO iteration count for reaching a comparable coarse-success regime.

9.3 Precision tradeoff

The advantage disappears when the threshold is tightened. Eureka achieved only 19.02% at 2 cm and 3.56% at 1 cm, versus 98.36% and 79.74% for vanilla. Mean terminal distance was 33.5 mm versus 7.6 mm.

9.4 Raw reward is not a valid comparison

The numerical scales of the two reward functions are fundamentally different. A larger total return for Eureka does not mean a better controller. External task metrics—terminal distance, thresholded success, speed, effort, robustness, and convergence time—are the correct comparison basis.

9.5 Reward correlation

The project reports a correlation of approximately 0.9995 between the Eureka reward and the oracle/human reward under the measured analysis. This suggests strong alignment at that measurement level, but it does not prove behavioral identity. Two highly correlated reward signals can still induce different optimal policies, which is exactly what occurred here.

10. Precision Failure Analysis

10.1 The central mechanism

Once the policy enters d<5 cm, the Eureka reward grants +100 regardless of whether the end-effector is 4.9 cm, 1 cm, or 1 mm from the target. The remaining distance-sensitive exponential term varies only on the order of 0–1, so it is tiny compared with the success bonus.

10.2 Reward landscape

The resulting incentive structure is approximately: outside 5 cm → very strong incentive to cross the boundary → +100 success reward → objective largely satisfied. There is comparatively little economic value in spending additional time refining the final centimeters.

10.3 Why more training did not solve it

The 1,500-iteration rerun was specifically useful because it tested whether the poor precision was simply under-training. The success bonus continued to dominate the total return. Around iteration 1522, the documented total reward was about 2,868.6, with approximately 2,854.5 attributed to the success bonus, while other terms were much smaller (roughly −7.2 for the velocity component and −0.3 for the time term in the reported snapshot).

10.4 Plateau behavior

The action standard deviation remained around the reported late-training range, while the success metric fluctuated rather than showing a sustained trend toward finer accuracy. This is consistent with convergence to a coarse objective rather than with a simple shortage of PPO iterations.

10.5 The 5 cm threshold problem

A 5 cm success threshold is a coarse notion of accuracy for many robotic applications. Reporting only P(d<5 cm) can make two controllers look nearly equivalent while hiding a very large difference in actual terminal error.

10.6 Visual marker versus RL objective

The visual green goal sphere has radius 0.05 m and therefore a 0.10 m diameter, but the visual geometry itself does not define the RL objective. The environment separately stores the target position and calculates d=||p_t−p_ee||. Shrinking the visual sphere alone would therefore not change Eureka's reward.

10.7 Better task specification

The review proposes a richer task description: move quickly while minimizing final Cartesian error, achieve millimeter- or sub-centimeter-level accuracy, avoid overshoot and oscillation, and use reasonable joint velocity and control effort. This is intended to make the desired behavior explicit to the reward-search process.

11. Domain Randomization Plan

11.1 Motivation

The next major stage is to make simulation less deterministic. Real hardware contains manufacturing tolerances, friction variation, actuator degradation, payload differences, sensor noise, external disturbances, and unmodeled dynamics.

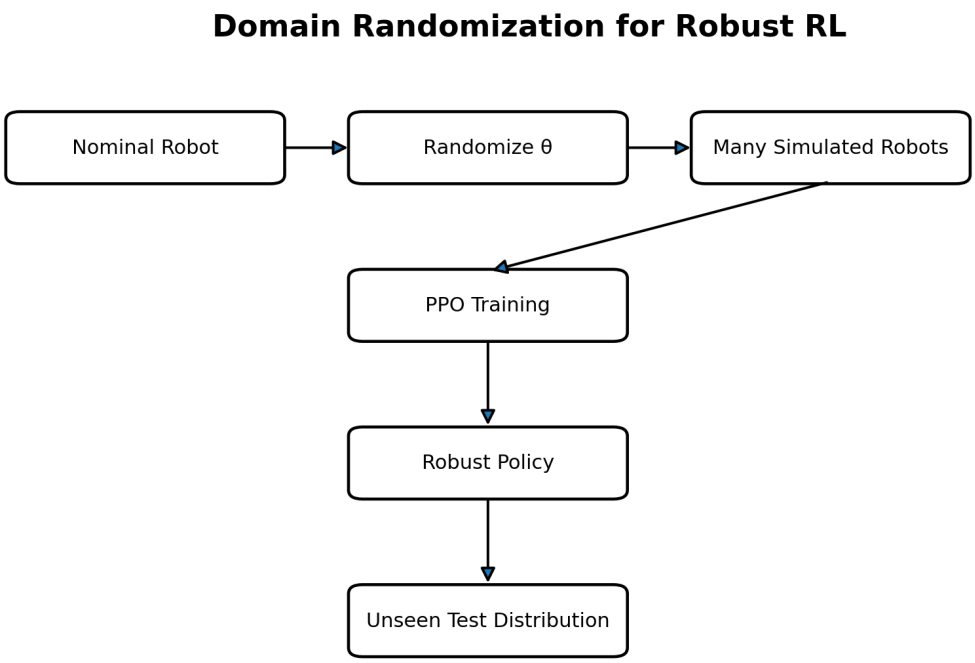


Figure 7. Proposed domain-randomized robust-learning pipeline.

11.2 Research question

The next question is whether Eureka can discover rewards that produce robust behavior across a family of possible robot and environment dynamics, rather than only one nominal simulated robot.

11.3 Joint friction randomization

Instead of fixing friction at one nominal value f■, the proposal samples friction from a distribution such as f~U(f_min,f_max). The documented example uses approximately ±20% variation during training.

11.4 Actuator degradation

The proposed actuator model introduces a persistent motor-effectiveness parameter α_i and applies it to commanded torque:

$$\tau_i = 20a_i, \qquad a_i \in [-1, 1], \qquad \tau_i \in [-20, 20]\ \text{Nm}$$

More specifically, the project proposes $\tau_{i,actual} = \alpha_i \tau_{i,commanded}$ with α_i sampled over a degradation range. An example $\alpha_i=0.8$ means the actuator produces only 80% of commanded torque.

11.5 External disturbances

External force can be represented as $F_{ext}=[F_x,F_y]$ with $F_x,F_y \sim U(-F_{max},F_{max})$. The point is to model a generic external disturbance rather than relying on an informal label such as “wind.”

11.6 Persistent versus instantaneous randomization

For physical properties such as friction and actuator effectiveness, sampling once per environment or episode is more meaningful than changing them every timestep. This exposes the policy to different persistent physical robots rather than a single robot whose properties teleport randomly from one step to the next.

11.7 Training versus unseen testing

The review proposes training on one uncertainty distribution and evaluating on a shifted, unseen distribution. The example training ranges include friction ±20% and motor degradation 0–20%, while the stronger unseen test can include friction ±30% and motor degradation 20–30%. This tests generalization rather than memorization of the training distribution.

12. Experimental Matrix and Scientific Controls

12.1 Proposed progression

The project is structured as a sequence in which each experiment answers a specific question.

Experiment	Environment	Reward	Purpose
A	Deterministic	Human baseline	Establish reference controller
B	Deterministic	Eureka	Test automated reward discovery
C	Domain randomized	Human + DR	Measure effect of DR on baseline
D	Domain randomized	Eureka + DR	Test Eureka under uncertainty
E	Domain randomized	Eureka + refined task	Test precision-aware reward discovery
F	Real / sim-to-real	DR policy	Transfer baseline
G	Real / sim-to-real	PINN-enhanced	Investigate physics-informed transfer

Table 4. Recommended experimental matrix documented in the review.

12.2 Scientific control variables

Across comparisons, the robot morphology, masses, link lengths, action dimensionality, observation dimensionality, PPO architecture, PPO hyperparameters, number of environments, simulation timestep, control frequency, target distribution, episode duration, evaluation procedure, randomization distribution, and evaluation seeds should remain constant wherever possible.

12.3 Primary independent variable

The intended primary difference should be the reward-generation mechanism. This is necessary for a defensible comparison between human reward engineering and Eureka-based reward discovery.

13. Robustness Metrics

13.1 Accuracy

Once domain randomization is introduced, accuracy alone is insufficient. The evaluation should include mean and median terminal error and a precision curve across 5 cm, 2 cm, 1 cm, 5 mm, 2 mm, and 1 mm.

13.2 Precision curve

A precision curve P(d<ε) gives a richer behavioral fingerprint than a single threshold. Two policies with similar 5 cm success can be fundamentally different if one collapses at 1 cm while the other retains high probability of fine accuracy.

13.3 Dynamic behavior

The review recommends settling time, overshoot, oscillation, maximum velocity, and related trajectory-level quantities.

13.4 Control cost

Control effort should be tracked explicitly, including a quantity such as the accumulated squared torque or action magnitude.

13.5 Training efficiency

Useful measures include PPO iterations, wall-clock time, number of Eureka reward candidates, failed candidates, and convergence behavior.

13.6 Multi-objective view

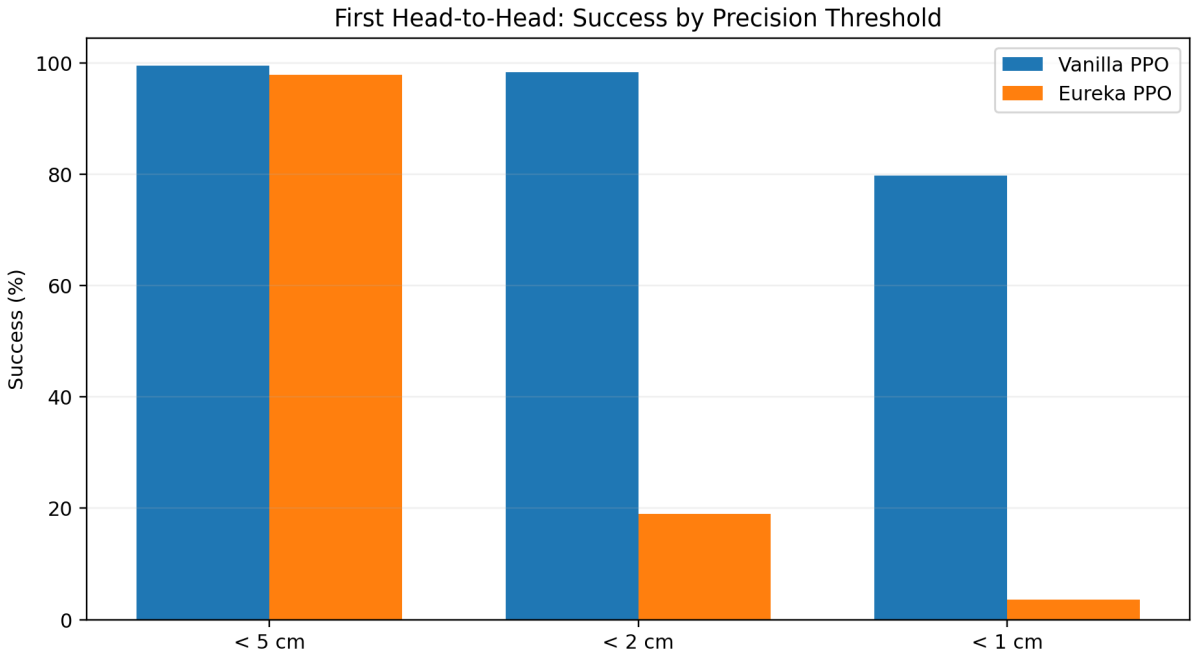


Figure 8. Precision thresholds reveal why a single success number is insufficient.

A conceptual multi-objective formulation

The review expresses the broader objective conceptually as a weighted combination of distance, time, velocity, effort, and overshoot. In abstract form:

```
J = w_d J_distance + w_t J_time + w_v J_velocity + w_τ J_effort + w_o J_overshoot.
```

14. Long-Term Sim-to-Real Direction and PINN Role

14.1 Ultimate objective

The long-term motivation is not simply to produce a good simulation policy. The goal is a policy sufficiently robust to remain useful when moved from an imperfect simulator to physical hardware.

14.2 Domain randomization as the robustness bridge

$$\max_{\pi} \mathbb{E}_{\theta \sim p(\theta)} [J(\pi; \theta)]$$

Equation 9. Conceptual robust-training objective over uncertain dynamics θ.

14.3 PINN is a separate research layer

The planned PINN is not intended to accelerate Eureka's reward search. Its role is associated with the later physics-informed sim-to-real stage. Eureka performs automated reward engineering; PPO performs policy optimization; domain randomization seeks robustness; physics-informed learning is subsequently investigated for sim-to-real transfer.

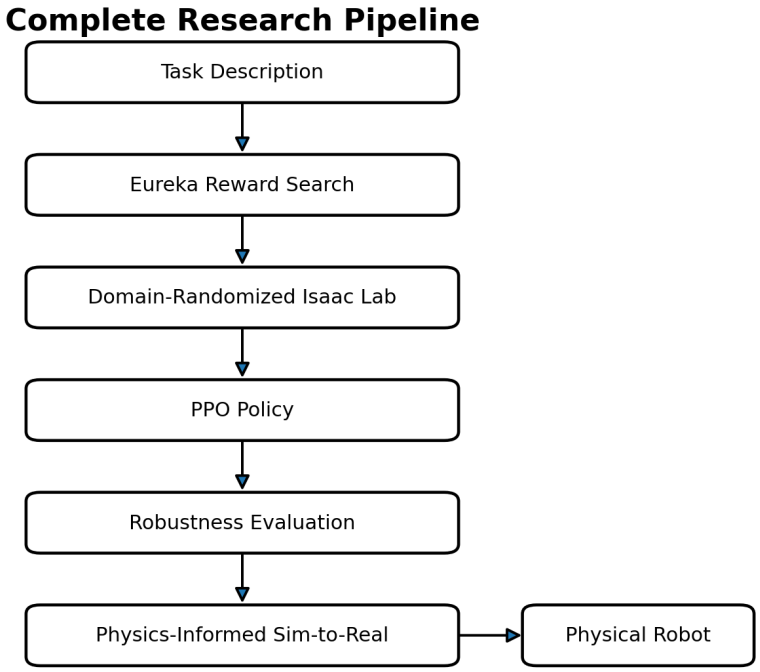


Figure 9. Full long-term pipeline from task description to physical robot.

14.4 Scientifically clean interpretation

The correct claim is not “PINN improves Eureka.” The intended separation is: Eureka automates reward discovery in simulation; domain randomization improves robustness to dynamics variation; and physics-informed learning is later investigated as a mechanism for improving transfer from simulation to real hardware.

15. What Has Been Established vs. What Remains Open

15.1 Established

• A 2-DOF planar torque-controlled reaching environment has been implemented.

• Forward kinematics and random target generation are implemented.

• The action interface uses two normalized torque actions scaled to ±20 Nm.

• The observation is eight-dimensional.

• The system runs at 120 Hz physics and 30 Hz policy control with 10-second episodes.

• 4,096 parallel Isaac Lab environments are used.

• A mathematically interpretable human reward and PPO baseline have been trained.

• A 5,000-iteration vanilla experiment and quantitative evaluation framework exist.

• Eureka has been integrated with the custom task and Gemini generated a valid reward.

• A 500-iteration Eureka experiment and 1,500-iteration rerun were evaluated.

• Eureka produced a functional reaching controller rapidly but with coarse precision.

• Longer training of the same Eureka reward did not materially close the precision gap.

15.2 Not yet established

• Eureka universally outperforms human reward engineering.

• Eureka produces more accurate robotic controllers.

• Eureka is more robust to environmental uncertainty.

• Eureka transfers better to real hardware.

• Domain randomization improves the Eureka policy relative to a controlled baseline.

• PINNs improve sim-to-real transfer for this particular system.

• Eureka-generated rewards generalize across different robots.

• The current 5 cm task should be treated as a universal definition of industrial robotic accuracy.

Scientific status: the work is a useful proof of concept, not a universal superiority claim. The strongest supported result is that automated reward discovery can rapidly generate a functional coarse-reaching controller, while the resulting behavior depends strongly on how the task objective is represented.

16. Final Research Roadmap

16.1 Immediate next step

Introduce domain randomization inspired by the DREUREKA direction described in the project and compare Vanilla PPO + DR against Eureka + DR under the same randomized physical conditions.

16.2 Precision-aware Eureka experiment

Before judging Eureka's intrinsic ability to discover precision-oriented rewards, strengthen the task specification so that accuracy is explicitly expressed at the millimeter/sub-centimeter level, while also mentioning overshoot, oscillation, velocity, and control effort.

16.3 Robustness evaluation

Use both in-distribution and shifted out-of-distribution dynamics. Report accuracy, precision thresholds, dynamic behavior, effort, and training efficiency rather than relying only on 5 cm success.

16.4 Sim-to-real stage

After robust simulation policies are established, investigate a physics-informed stage. The PINN should be treated as a mechanism for representing or exploiting physical structure during transfer, not as a component that changes the Eureka reward-search mechanism.

16.5 Final research question

The mature research question becomes: can automated reward design produce robust and genuinely precise robotic controllers under uncertain dynamics, and can physics-informed methods subsequently improve their transfer from simulation to physical hardware?

16.6 Three research layers

The project therefore has three coherent layers: (1) automated reward design, (2) robust learning, and (3) sim-to-real physics, with PPO acting as the policy-learning mechanism throughout the reinforcement-learning stage.

Appendix A. Consolidated Mathematical Reference

A.1 Manipulator dynamics

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) + F(\dot{q}) = \tau + \tau_{\text{ext}}$$

A.2 Forward kinematics

$$x_{ee} = l_1\cos q_1 + l_2\cos(q_1+q_2), \qquad y_{ee} = l_1\sin q_1 + l_2\sin(q_1+q_2)$$

A.3 Target generation

$$r \sim U(0.20, 0.85), \quad \theta \sim U(0, 2\pi), \quad x_t = r\cos\theta, \quad y_t = r\sin\theta$$

A.4 Cartesian error

$$e_t = p_t - p_{ee,t}, \qquad d_t = \|e_t\|_2 = \sqrt{(x_t - x_{ee,t})^2 + (y_t - y_{ee,t})^2}$$

A.5 Observation vector

$$s_t = [q_1, q_2, \dot{q}_1, \dot{q}_2, x_t, y_t, x_t - x_{ee}, y_t - y_{ee}]$$

A.6 Action-to-torque mapping

$$\tau_i = 20a_i, \qquad a_i \in [-1, 1], \qquad \tau_i \in [-20, 20]\ \text{Nm}$$

A.7 Human reward

$$R_{\text{vanilla}} = -d + 5\,\mathbb{I}(d < 0.05) - 0.005\|a\|_2^2$$

A.8 Eureka reward

$$R_{\text{Eureka}} = e^{-10d} + 100\,\mathbb{I}(d < 0.05) - 0.01\|\dot{q}\|_2^2 - 0.1$$

A.9 PPO objective

$$\pi^* = \arg\max_{\pi} \mathbb{E}_{\pi}\left[\sum_t \gamma^t R_t\right]$$

A.10 Robust domain-randomized objective

$$\max_{\pi}\ \mathbb{E}_{\theta\sim p(\theta)}\left[J(\pi;\theta)\right]$$

The formulas above are the mathematical backbone of the documented system. Together they form the chain q → forward kinematics → end-effector position → Cartesian error → reward → PPO objective, while domain randomization extends the optimization from one nominal dynamics instance to an expectation over uncertain dynamics.

Appendix B. Consolidated Configuration and Results

System item	Documented configuration
Robot	2-DOF planar serial arm
Link lengths	0.5 m, 0.5 m
Link masses	1.0 kg, 0.5 kg
Joints	2 revolute joints about Z
End-effector	Tip of link 2
Joint position range	≈ [–π, π]
Torque range	±20 Nm
Velocity range	≈ ±5 rad/s
Observation dimension	8
Action dimension	2 normalized actions
Physics frequency	120 Hz
Control frequency	30 Hz
Decimation	4
Episode duration	10 s
Policy decisions / episode	≈300
Physics steps / episode	≈1200
Parallel environments	4096
Rollout transitions	4096 × 24 = 98,304
PPO network	8 → 256 → 128 → 64
Activation	ELU
Initial action std.	1.0
Learning rate	1e–3
γ / λ	0.99 / 0.95
Clip / entropy	0.2 / 0.005
Desired KL	0.01
Max grad norm	1.0

Table B1. Consolidated system and PPO configuration.

Metric	Vanilla	Eureka 500	Eureka 1500
Iterations	5000	500	≈1500
<5 cm	99.49%	97.90%	95.3%
<2 cm	98.36%	19.02%	18.1%
<1 cm	79.74%	3.56%	5.3%
Mean error	7.6 mm	33.5 mm	45.5 mm
Median error	6.3 mm	34.6 mm	33.3 mm

Metric	Vanilla	Eureka 500	Eureka 1500
Mean velocity	0.315 rad/s	4.844 rad/s	4.79 rad/s
Mean effort	≈0.218	≈2.198	≈2.54

Table B2. Consolidated evaluation results.

Final Synthesis

The first stage of the project has established a clean experimental foundation for studying automated reward design in robotics. A simple but nontrivial torque-controlled 2-DOF arm is simulated at high parallel scale; PPO learns the low-level controller; a human reward provides a strong precision baseline; Eureka replaces manual reward construction with an LLM-driven search process; and the resulting policies are evaluated using external task metrics.

The most important empirical pattern is a clear speed–precision tradeoff. Eureka rapidly discovered a reward that made PPO good at entering the 5 cm success region. The human reward took substantially more PPO iterations but produced much better centimeter and millimeter-level precision. Analysis of the reward equations explains why: Eureka's +100 success bonus dwarfed its residual distance shaping once the boundary was crossed, so additional fine positioning had little reward value. The later 1,500-iteration experiment showed that this was an objective-design issue rather than simply an iteration budget issue.

The project therefore moves naturally into robust RL: domain-randomize friction, actuator effectiveness, and external disturbances; compare vanilla PPO + DR against Eureka + DR under controlled conditions; test shifted uncertainty distributions; and evaluate not just coarse success but precision, dynamics, effort, and robustness. Only after this robust simulation stage does the physics-informed PINN layer become relevant, as a future mechanism for sim-to-real transfer.

Central finding: Eureka changes the objective; PPO changes the policy. The first Eureka reward was a rapid coarse-reaching solution, not a precision-oriented solution. The scientific opportunity now is to determine whether richer task specifications and domain-randomized training can turn automated reward discovery into a mechanism for robust, precise robotic control, followed by a physics-informed route toward real-hardware transfer.

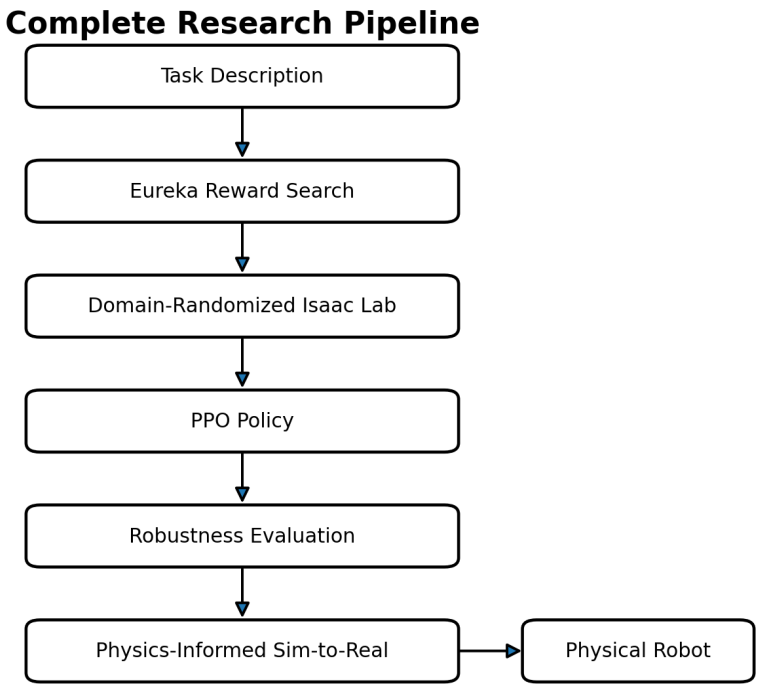


Figure 10. End-to-end research direction: Automated Reward Design → Robust RL → Sim-to-Real Robotics.